



National  
Qualifications  
2025

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## 2025 Mathematics

### Higher - Paper 2

## Question Paper Finalised Marking Instructions

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## General marking principles for Higher Mathematics

Always apply these general principles. Use them in conjunction with the detailed marking instructions, which identify the key features required in candidates' responses.

For each question, the marking instructions are generally in two sections:

- *generic scheme* – this indicates why each mark is awarded
- *illustrative scheme* – this covers methods which are commonly seen throughout the marking

In general, you should use the illustrative scheme. Only use the generic scheme where a candidate has used a method not covered in the illustrative scheme.

- Always use positive marking. This means candidates accumulate marks for the demonstration of relevant skills, knowledge and understanding; marks are not deducted for errors or omissions.
- If you are uncertain how to assess a specific candidate response because it is not covered by the general marking principles or the detailed marking instructions, you must seek guidance from your team leader.
- One mark is available for each •. There are no half marks.
- If a candidate's response contains an error, all working subsequent to this error must still be marked. Only award marks if the level of difficulty in their working is similar to the level of difficulty in the illustrative scheme.
- Only award full marks where the solution contains appropriate working. A correct answer with no working receives no mark, unless specifically mentioned in the marking instructions.
- Candidates may use any mathematically correct method to answer questions, except in cases where a particular method is specified or excluded.
- If an error is trivial, casual or insignificant, for example  $6 \times 6 = 12$ , candidates lose the opportunity to gain a mark, except for instances such as the second example in point (h) below.
- If a candidate makes a transcription error (question paper to script or within script), they lose the opportunity to gain the next process mark, for example

This is a transcription error and so the mark is not awarded.

This is no longer a solution of a quadratic equation, so the mark is not awarded.

$$x^2 + 5x + 7 = 9x + 4$$

$$x - 4x + 3 = 0$$

$$x = 1$$

The following example is an exception to the above

This error is not treated as a transcription error, as the candidate deals with the intended quadratic equation. The candidate has been given the benefit of the doubt and all marks awarded.

$$x^2 + 5x + 7 = 9x + 4$$

$$x - 4x + 3 = 0$$

$$(x - 3)(x - 1) = 0$$

$$x = 1 \text{ or } 3$$

(i) **Horizontal/vertical marking**

If a question results in two pairs of solutions, apply the following technique, but only if indicated in the detailed marking instructions for the question.

Example:

$$\begin{array}{cc} \bullet^5 & \bullet^6 \\ \bullet^5 & x = 2 \quad x = -4 \\ \bullet^6 & y = 5 \quad y = -7 \end{array}$$

Horizontal:  $\bullet^5 x = 2$  and  $x = -4$     Vertical:  $\bullet^5 x = 2$  and  $y = 5$   
 $\bullet^6 y = 5$  and  $y = -7$      $\bullet^6 x = -4$  and  $y = -7$

You must choose whichever method benefits the candidate, **not** a combination of both.

(j) In final answers, candidates should simplify numerical values as far as possible unless specifically mentioned in the detailed marking instruction. For example

$$\begin{array}{ll} \frac{15}{12} \text{ must be simplified to } \frac{5}{4} \text{ or } 1\frac{1}{4} & \frac{43}{1} \text{ must be simplified to } 43 \\ \frac{15}{0.3} \text{ must be simplified to } 50 & \frac{4\cancel{5}}{3} \text{ must be simplified to } \frac{4}{15} \\ \sqrt{64} \text{ must be simplified to } 8^* & \end{array}$$

\*The square root of perfect squares up to and including 144 must be known.

- (k) Commonly Observed Responses (COR) are shown in the marking instructions to help mark common and/or non-routine solutions. CORs may also be used as a guide when marking similar non-routine candidate responses.
- (l) Do not penalise candidates for any of the following, unless specifically mentioned in the detailed marking instructions:
- working subsequent to a correct answer
  - correct working in the wrong part of a question
  - legitimate variations in numerical answers/algebraic expressions, for example angles in degrees rounded to nearest degree
  - omission of units
  - bad form (bad form only becomes bad form if subsequent working is correct), for example  
 $(x^3 + 2x^2 + 3x + 2)(2x + 1)$  written as  
 $(x^3 + 2x^2 + 3x + 2) \times 2x + 1$   
 $= 2x^4 + 5x^3 + 8x^2 + 7x + 2$  gains full credit
  - repeated error within a question, but not between questions or papers
- (m) In any 'Show that...' question, where candidates have to arrive at a required result, the last mark is not awarded as a follow-through from a previous error, unless specified in the detailed marking instructions.

- (n) You must check all working carefully, even where a fundamental misunderstanding is apparent early in a candidate's response. You may still be able to award marks later in the question so you must refer continually to the marking instructions. The appearance of the correct answer does not necessarily indicate that you can award all the available marks to a candidate.
- (o) You should mark legible scored-out working that has not been replaced. However, if the scored-out working has been replaced, you must only mark the replacement working.
- (p) If candidates make multiple attempts using the same strategy and do not identify their final answer, mark all attempts and award the lowest mark. If candidates try different valid strategies, apply the above rule to attempts within each strategy and then award the highest mark.

For example:

|                                                                    |                                                                    |
|--------------------------------------------------------------------|--------------------------------------------------------------------|
| Strategy 1 attempt 1 is worth 3 marks.                             | Strategy 2 attempt 1 is worth 1 mark.                              |
| Strategy 1 attempt 2 is worth 4 marks.                             | Strategy 2 attempt 2 is worth 5 marks.                             |
| From the attempts using strategy 1, the resultant mark would be 3. | From the attempts using strategy 2, the resultant mark would be 1. |

In this case, award 3 marks.

Note: Marking from Image (MFI) annotation change from 2025

A double cross-tick is used to indicate correct working which is irrelevant or insufficient to score any marks. In MFI marking instructions prior to 2025 this was shown as  $\ddot{u}_2$  or  $\ddot{u}2$ .

From 2025, the double cross-tick will no longer be used in MFI. A cross or omission symbol will be used instead.

# Marking instructions for each question

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                 |     |  | Generic scheme                                                                                                                                        | Illustrative scheme                                                                                                                                                                                                                                    | Max mark                                                                                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.                                                                                                                                                                                                                                                                                                                                                                                                                                                       | (a) |  | <p>•<sup>1</sup> find gradient of AC</p> <p>•<sup>2</sup> use property of perpendicular lines</p> <p>•<sup>3</sup> determine equation of altitude</p> | <p>•<sup>1</sup> <math>-\frac{1}{3}</math></p> <p>OR</p> <p><math>-\frac{10}{30}</math></p> <p>•<sup>2</sup> 3</p> <p>OR</p> <p><math>\frac{30}{10}</math></p> <p>•<sup>3</sup> <math>y = 3x - 7</math></p>                                            | 3                                                                                                                                                          |
| <b>Notes:</b> <ol style="list-style-type: none"> <li>•<sup>3</sup> is only available to candidates who find and use a perpendicular gradient.</li> <li>The gradient of the altitude must appear in fully simplified form at the •<sup>2</sup> or •<sup>3</sup> stage for •<sup>3</sup> to be awarded. See Candidate A.</li> <li>At •<sup>3</sup>, accept any arrangement of a candidate's equation where constant terms have been simplified.</li> </ol> |     |  |                                                                                                                                                       |                                                                                                                                                                                                                                                        |                                                                                                                                                            |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                                                                                                                                                                                      |     |  |                                                                                                                                                       |                                                                                                                                                                                                                                                        |                                                                                                                                                            |
| <b>Candidate A - not simplifying the gradient</b><br>$10y = 30x - 70$                                                                                                                                                                                                                                                                                                                                                                                    |     |  | <p>•<sup>3</sup> ✗</p>                                                                                                                                | <p><b>Candidate B - <math>m = m_{\text{perp}}</math></b></p> <p><math>m = -\frac{1}{3} = 3</math></p> <p><math>y = 3x - 7</math></p> <p>However</p> <p><math>m = -\frac{1}{3}</math></p> <p><math>= 3</math></p> <p><u><math>y = 3x - 7</math></u></p> | <p>•<sup>1</sup> ✓ •<sup>2</sup> ✗</p> <p>•<sup>3</sup> ✓<sub>1</sub> - BOD</p> <p>•<sup>1</sup> ✓</p> <p>•<sup>2</sup> ✓ - BOD</p> <p>•<sup>3</sup> ✓</p> |



| Question                                                                                                                                                                                    |  |  | Generic scheme                                                                                                                                                                                                                                                   | Illustrative scheme                                                                                                                                                                                                                                                          | Max mark |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 2.                                                                                                                                                                                          |  |  | <b>Method 1</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> identify common factor</li> <li>•<sup>2</sup> complete the square</li> <li>•<sup>3</sup> process for <math>r</math> and write in required form</li> </ul>                                   | <b>Method 1</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>2(x^2 + 8x \dots</math><br/>stated or implied by •<sup>2</sup></li> <li>•<sup>2</sup> <math>2(x + 4)^2 \dots</math></li> <li>•<sup>3</sup> <math>2(x + 4)^2 - 27</math></li> </ul>                | 3        |
|                                                                                                                                                                                             |  |  | <b>Method 2</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> expand completed square form</li> <li>•<sup>2</sup> equate coefficients</li> <li>•<sup>3</sup> process for <math>q</math> and <math>r</math> and write in required form</li> </ul>          | <b>Method 2</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>px^2 + 2pqx + pq^2 + r</math> stated or implied by •<sup>2</sup></li> <li>•<sup>2</sup> <math>p = 2, 2pq = 16, pq^2 + r = 5</math></li> <li>•<sup>3</sup> <math>2(x + 4)^2 - 27</math></li> </ul> |          |
| <b>Notes:</b>                                                                                                                                                                               |  |  |                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                              |          |
| 1. $2(x + 4)^2 - 27$ with no working gains • <sup>1</sup> and • <sup>2</sup> only. However, see Candidate E.<br>2. Do not penalise candidates who do not work with $p, q$ and $r$ .         |  |  |                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                              |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                         |  |  |                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                              |          |
| <b>Candidate A</b><br>$2(x^2 + 8) + 5$<br>$2((x + 4)^2 - 16) + 5$ • <sup>1</sup> ✓ • <sup>2</sup> ✓<br>$2(x + 4)^2 - 27$ • <sup>3</sup> ✓<br>See exception to general marking principle (h) |  |  | <b>Candidate B - not using required form</b><br>$px^2 + 2pqx + pq^2 + r$ • <sup>1</sup> ✓<br>$p = 2, 2pq = 16, pq^2 + r = 5$ • <sup>2</sup> ✓<br>$q = 4, r = -27$ • <sup>3</sup> ^<br><div>•<sup>3</sup> is lost as answer is not in completed square form</div> |                                                                                                                                                                                                                                                                              |          |
| <b>Candidate C</b><br>$2(x^2 + 16x) + 5$ • <sup>1</sup> ✗<br>$2((x + 8)^2 - 64) + 5$ • <sup>2</sup> ✓ <sub>1</sub><br>$2(x + 8)^2 - 123$ • <sup>3</sup> ✓ <sub>1</sub>                      |  |  | <b>Candidate D</b><br>$2((x + 8)^2 - 64) + 5$ • <sup>1</sup> ✗ • <sup>2</sup> ✗<br>$2(x + 8)^2 - 123$ • <sup>3</sup> ✓ <sub>1</sub>                                                                                                                              |                                                                                                                                                                                                                                                                              |          |
| <b>Candidate E</b><br>$2(x + 4)^2 - 27$ • <sup>1</sup> ✓ • <sup>2</sup> ✓<br>Check:<br>$= 2(x^2 + 8x + 16) - 27$<br>$= 2x^2 + 16x + 32 - 27$<br>$= 2x^2 + 16x + 5$ • <sup>3</sup> ✓         |  |  | <b>Candidate F</b><br>$2(x + 4)^2 = 2x^2 + 16x + 32$ • <sup>1</sup> ✓ • <sup>2</sup> ✓<br>$\therefore 2(x + 4)^2 - 27 = 2x^2 + 16x + 5$ • <sup>3</sup> ✓                                                                                                         |                                                                                                                                                                                                                                                                              |          |





| Question |  |  | Generic scheme                                                                                                                                                                                                                                                      | Illustrative scheme                                                                                                                                                                                                                                                                                                                                   | Max mark |
|----------|--|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 4.       |  |  | <b>Method 1</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> equate composite function to <math>x</math></li> <li>•<sup>2</sup> write <math>g(g^{-1}(x))</math> in terms of <math>g^{-1}(x)</math></li> <li>•<sup>3</sup> state inverse function</li> </ul> | <b>Method 1</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>g(g^{-1}(x)) = x</math></li> <li>•<sup>2</sup> <math>(g^{-1}(x) - 4)^3 = x</math></li> <li>•<sup>3</sup> <math>g^{-1}(x) = \sqrt[3]{x} + 4</math></li> </ul>                                                                                                               | 3        |
|          |  |  | <b>Method 2</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> write as <math>y = (x - 4)^3</math> and start to rearrange</li> <li>•<sup>2</sup> complete rearrangement</li> <li>•<sup>3</sup> state inverse function</li> </ul>                              | <b>Method 2</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>y = g(x) \Rightarrow x = g^{-1}(y)</math><br/><math>\sqrt[3]{y} = x - 4</math></li> <li>•<sup>2</sup> <math>x = 4 + \sqrt[3]{y}</math></li> <li>•<sup>3</sup> <math>g^{-1}(y) = \sqrt[3]{y} + 4</math><br/><math>\Rightarrow g^{-1}(x) = \sqrt[3]{x} + 4</math></li> </ul> |          |

#### Notes:

1. In method 1, accept  $x = (g^{-1}(x) - 4)^3$  for •<sup>1</sup> and •<sup>2</sup>.
2. In method 2, accept ' $\sqrt[3]{y} = x - 4$ ' without reference to  $y = g(x) \Rightarrow x = g^{-1}(y)$  at •<sup>1</sup>.
3. In method 2, accept  $g^{-1}(x) = \sqrt[3]{x} + 4$  without reference to  $g^{-1}(y)$  at •<sup>3</sup>.
4. In method 2, beware of candidates with working where each line is not mathematically equivalent. See Candidates A, B and C.
5. At •<sup>3</sup> stage, accept  $g^{-1}$  written in terms of any dummy variable. For example  $g^{-1}(y) = \sqrt[3]{y} + 4$ .
6.  $y = \sqrt[3]{x} + 4$  does not gain •<sup>3</sup>.
7.  $g^{-1}(x) = \sqrt[3]{x} + 4$  with no working gains 3/3.
8. In method 2, where candidates make multiple algebraic errors at the •<sup>2</sup> stage, •<sup>3</sup> is still available.
9. Marks should only be awarded for using a valid strategy to find the inverse of  $g(x)$ .

#### Commonly Observed Responses:

##### Candidate A

$$g(x) = (x - 4)^3$$

$$y = (x - 4)^3 \quad \text{---}$$

$$x = \sqrt[3]{y} + 4$$

$$y = \sqrt[3]{x} + 4 \quad \text{---}$$

$$g^{-1}(x) = \sqrt[3]{x} + 4$$

•<sup>1</sup> ✓ •<sup>2</sup> ✓  
•<sup>3</sup> ✗

##### Candidate B

$$g(x) = (x - 4)^3$$

$$y = (x - 4)^3 \quad \text{---}$$

$$x = (y - 4)^3 \quad \text{---}$$

$$y = \sqrt[3]{x} + 4$$

$$g^{-1}(x) = \sqrt[3]{x} + 4$$

•<sup>1</sup> ✗  
•<sup>2</sup> ✓<sub>1</sub>  
•<sup>3</sup> ✓<sub>1</sub>

| Question                                          |             | Generic scheme                | Illustrative scheme                                | Max mark         |
|---------------------------------------------------|-------------|-------------------------------|----------------------------------------------------|------------------|
| 4.                                                | (continued) |                               |                                                    |                  |
| <b>Candidate C</b>                                |             |                               | <b>Candidate D - Method 1</b>                      |                  |
| $x = (g(x) - 4)^3$                                |             | • <sup>1</sup> ✗              | $g(g^{-1}(x)) = (g^{-1}(x) - 4)^3$                 | • <sup>2</sup> ✓ |
| $g(x) = \sqrt[3]{x} + 4$                          |             | • <sup>2</sup> ✓ <sub>1</sub> | $x = (g^{-1}(x) - 4)^3$                            | • <sup>1</sup> ✓ |
| $g^{-1}(x) = \sqrt[3]{x} + 4$                     |             | • <sup>3</sup> ✓ <sub>1</sub> | $g^{-1}(x) = \sqrt[3]{x} + 4$                      | • <sup>3</sup> ✓ |
| <b>Candidate E - BEWARE of incorrect notation</b> |             |                               | <b>Candidate F</b>                                 |                  |
| $g'(x) = \dots$                                   |             | • <sup>3</sup> ✗              | $x \rightarrow x - 4 \rightarrow (x - 4)^3 = g(x)$ |                  |
| $f^{-1}(x) = \dots$                               |             | • <sup>3</sup> ✗              | $-4 \rightarrow^3$                                 |                  |
|                                                   |             |                               | $\therefore \sqrt[3]{\phantom{x}} \rightarrow +4$  | • <sup>1</sup> ✓ |
|                                                   |             |                               | $\sqrt[3]{x} + 4$                                  | • <sup>2</sup> ✓ |
|                                                   |             |                               | $g^{-1}(x) = \sqrt[3]{x} + 4$                      | • <sup>3</sup> ✓ |

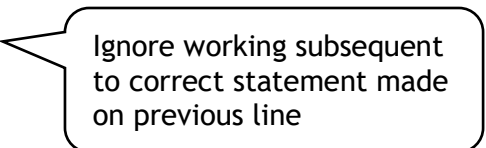
| Question |     |  | Generic scheme                                                                                                                                                                                                                                                           | Illustrative scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Max mark |
|----------|-----|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 5.       | (a) |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> find an appropriate vector eg <math>\overrightarrow{AB}</math></li> <li>•<sup>2</sup> find a second vector eg <math>\overrightarrow{BC}</math> AND compare</li> <li>•<sup>3</sup> appropriate conclusion</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> eg <math>\overrightarrow{AB} = \begin{pmatrix} 9 \\ -3 \\ 6 \end{pmatrix}</math></li> <li>•<sup>2</sup> eg <math>\overrightarrow{BC} = \begin{pmatrix} 6 \\ -2 \\ 4 \end{pmatrix} \therefore \overrightarrow{AB} = \frac{3}{2}\overrightarrow{BC}</math></li> <li>•<sup>3</sup> ... <math>\Rightarrow</math> AB is parallel to BC (common direction)</li> </ul> <p>AND</p> <p>B is a common point<br/> <math>\Rightarrow</math> A, B and C are collinear.</p> | 3        |

#### Notes:

1. Do not penalise inconsistent vector notation (for example lack of arrows or brackets).
2. If no comparison of vectors or the trivial comparison ' $\overrightarrow{AB} = \overrightarrow{BC}$ ' is made at •<sup>2</sup>, then •<sup>3</sup> is not available.
3. •<sup>3</sup> can only be awarded if a candidate has stated 'parallel', 'common point' and 'collinear'.
4. Candidates who state that 'points are parallel' or 'vectors are collinear' or 'parallel and share a common point  $\Rightarrow$  collinear' do not gain •<sup>3</sup>. There must be a reference to the points.
5. Do not accept ' $a, b$  and  $c$  are collinear' at •<sup>3</sup>.

#### Commonly Observed Responses:

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Candidate A - missing labels</b></p> $\begin{pmatrix} 9 \\ -3 \\ 6 \end{pmatrix}$ <p>•<sup>1</sup> ^</p> $\begin{pmatrix} 6 \\ -2 \\ 4 \end{pmatrix} \therefore \overrightarrow{AB} = \frac{3}{2}\overrightarrow{BC}$ <p>•<sup>2</sup> ✓<sub>1</sub></p> <p>Missing labels at •<sup>2</sup> is a repeated error</p> <p><math>\Rightarrow</math> AB is parallel to BC<br/> and B is a common point<br/> <math>\Rightarrow</math> A, B and C are collinear</p> <p>•<sup>3</sup> ✓<sub>1</sub></p> | <p><b>Candidate B</b></p> $\overrightarrow{AB} = \begin{pmatrix} 9 \\ -3 \\ 6 \end{pmatrix}$ <p>•<sup>1</sup> ✓</p> $\overrightarrow{BC} = \begin{pmatrix} 6 \\ -2 \\ 4 \end{pmatrix}$ $\begin{pmatrix} 9 \\ -3 \\ 6 \end{pmatrix} = 3 \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} \text{ AND } \begin{pmatrix} 6 \\ -2 \\ 4 \end{pmatrix} = 2 \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$ <p>•<sup>2</sup> ✓</p> <p><math>\therefore \overrightarrow{AB} = \frac{2}{3}\overrightarrow{BC}</math></p> <p>Ignore working subsequent to correct statement made on previous line</p> <p><math>\Rightarrow</math> AB is parallel to BC<br/> and B is a common point<br/> <math>\Rightarrow</math> A, B and C are collinear</p> <p>•<sup>3</sup> ✓</p> |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

| Question                                                                                                                                                                                                                                  |                                                         |                                                                                     | Generic scheme                                   | Illustrative scheme                                     | Max mark                      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------|---------------------------------------------------------|-------------------------------|
| 5.                                                                                                                                                                                                                                        | (b)                                                     |                                                                                     | • <sup>4</sup> state ratio                       | • <sup>4</sup> 3:2                                      | 1                             |
| <b>Notes:</b>                                                                                                                                                                                                                             |                                                         |                                                                                     |                                                  |                                                         |                               |
| 6. Answers in (b) must be consistent with the components of the vectors in (a) or the comparison of the vectors in (a). See Candidates C and D. However, award • <sup>4</sup> for '3:2' with no working.                                  |                                                         |                                                                                     |                                                  |                                                         |                               |
| 7. In this question, the answer for • <sup>4</sup> must be stated explicitly in part (b).                                                                                                                                                 |                                                         |                                                                                     |                                                  |                                                         |                               |
| 8. The only acceptable variations for • <sup>4</sup> must be related explicitly to AB and BC. For $\frac{BC}{AB} = \frac{2}{3}$ , $\frac{AB}{BC} = \frac{3}{2}$ or BC:AB = 2:3 stated in part (b) award • <sup>4</sup> . See Candidate E. |                                                         |                                                                                     |                                                  |                                                         |                               |
| 9. Accept unitary ratios for • <sup>4</sup> , for example $\frac{3}{2}:1$ or $1:\frac{2}{3}$ .                                                                                                                                            |                                                         |                                                                                     |                                                  |                                                         |                               |
| 10. Where candidates state multiple ratios which are not equivalent, award 0/1.                                                                                                                                                           |                                                         |                                                                                     |                                                  |                                                         |                               |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                       |                                                         |                                                                                     |                                                  |                                                         |                               |
| <b>Candidate C - using components of vectors</b>                                                                                                                                                                                          |                                                         |                                                                                     | <b>Candidate D - using comparison of vectors</b> |                                                         |                               |
| (a)                                                                                                                                                                                                                                       | $\vec{AB} = \begin{pmatrix} 9 \\ -3 \\ 6 \end{pmatrix}$ | • <sup>1</sup> ✓                                                                    | (a)                                              | $\vec{AB} = \begin{pmatrix} 9 \\ -3 \\ 6 \end{pmatrix}$ | • <sup>1</sup> ✓              |
|                                                                                                                                                                                                                                           | $\vec{BC} = \begin{pmatrix} 6 \\ -2 \\ 4 \end{pmatrix}$ |                                                                                     |                                                  | $\vec{BC} = \begin{pmatrix} 6 \\ -2 \\ 4 \end{pmatrix}$ |                               |
|                                                                                                                                                                                                                                           | $\vec{BC} = \frac{3}{2}\vec{AB}$                        | • <sup>2</sup> ✗                                                                    |                                                  | $\vec{BC} = \frac{3}{2}\vec{AB}$                        | • <sup>2</sup> ✗              |
| (b)                                                                                                                                                                                                                                       | 3:2                                                     | • <sup>4</sup> ✓                                                                    | (b)                                              | 2:3                                                     | • <sup>4</sup> ✓ <sub>1</sub> |
| <b>Candidate E - acceptable variation</b>                                                                                                                                                                                                 |                                                         |                                                                                     | <b>Candidate F - trivial ratio</b>               |                                                         |                               |
|                                                                                                                                                                                                                                           | $\frac{AB}{BC} = \frac{3}{2}$                           | • <sup>4</sup> ✓                                                                    |                                                  | Ratio = 1:1                                             | • <sup>4</sup> ✗              |
|                                                                                                                                                                                                                                           | Ratio = 2:3                                             |  |                                                  |                                                         |                               |

| Question |     |  | Generic scheme                                                                                                                                                                                                                                                          | Illustrative scheme                                                                                                                                                                                                                                                                                                                                               | Max mark |
|----------|-----|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 6.       | (a) |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> use compound angle formula</li> <li>•<sup>2</sup> compare coefficients</li> <li>•<sup>3</sup> process for <math>k</math></li> <li>•<sup>4</sup> process for <math>a</math> and express in required form</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>k \cos x \cos a - k \sin x \sin a</math> <b>stated explicitly</b></li> <li>•<sup>2</sup> <math>k \cos a = 5</math> and <math>k \sin a = 9</math> <b>stated explicitly</b></li> <li>•<sup>3</sup> <math>\sqrt{106}</math></li> <li>•<sup>4</sup> <math>\sqrt{106} \cos(x + 1.06\dots)</math></li> </ul> | 4        |

#### Notes:

1. Accept  $k(\cos x \cos a - \sin x \sin a)$  for •<sup>1</sup>.
2. Treat  $k \cos x \cos a - \sin x \sin a$  as bad form only if the equations at the •<sup>2</sup> stage both contain  $k$ .
3.  $\sqrt{106} \cos x \cos a - \sqrt{106} \sin x \sin a$  or  $\sqrt{106}(\cos x \cos a - \sin x \sin a)$  are acceptable for •<sup>1</sup> and •<sup>3</sup>.
4. •<sup>2</sup> is not available for  $k \cos x = 5$  and  $k \sin x = 9$ , however •<sup>4</sup> may still be gained. See Candidate E.
5. Accept  $-k \sin a = -9$  and  $k \cos a = 5$  for •<sup>2</sup>.
6. •<sup>4</sup> is not available for a value of  $a$  given in degrees.
7. Accept values of  $a$  which round to 1.1.
8. Candidates may state and use any form of the wave function for •<sup>1</sup>, •<sup>2</sup> and •<sup>3</sup>. However, •<sup>4</sup> is only available if the wave is interpreted in the form  $k \cos(x + a)$ .
9. Evidence for •<sup>4</sup> may not appear until part (b) and must appear by the •<sup>5</sup> stage.

#### Commonly Observed Responses:

|                                                                                                                                                                                                                                               |  |  |                                                                                                                                              |                                                                                                                                                                                                                               |  |  |                                                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Candidate A</b><br>$\sqrt{106} \cos a = 5$<br>$\sqrt{106} \sin a = 9$<br>$\tan a = \frac{9}{5}$<br>$a = 1.06\dots$<br>$\sqrt{106} \cos(x + 1.06\dots)$                                                                                     |  |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> ^</li> <li>•<sup>2</sup> ✓</li> <li>•<sup>3</sup> ✓</li> <li>•<sup>4</sup> ✓</li> </ul> | <b>Candidate B</b><br>$k \cos x \cos a - k \sin x \sin a$<br>$\cos a = 5$<br>$\sin a = 9$<br>$\tan a = \frac{9}{5}$<br>$a = 1.06\dots$ (Not consistent with equations at • <sup>2</sup> )<br>$\sqrt{106} \cos(x + 1.06\dots)$ |  |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> ✓</li> <li>•<sup>2</sup> ✗</li> <li>•<sup>3</sup> ✓</li> <li>•<sup>4</sup> ✗</li> </ul>             |
| <b>Candidate C</b><br>$\cos x \cos a - \sin x \sin a$<br>$\cos a = 5$<br>$\sin a = 9$<br>$k = \sqrt{106}$<br>$\tan a = \frac{9}{5}$<br>$a = 1.06\dots$ (Not consistent with equations at • <sup>2</sup> )<br>$\sqrt{106} \cos(x + 1.06\dots)$ |  |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> ✗</li> <li>•<sup>2</sup> ✗</li> <li>•<sup>3</sup> ✓</li> <li>•<sup>4</sup> ✗</li> </ul> | <b>Candidate D - errors at •<sup>2</sup></b><br>$k \cos x \cos a - k \sin x \sin a$<br>$k \cos a = 9$<br>$k \sin a = 5$<br>$\tan a = \frac{5}{9}$<br>$a = 0.507\dots$<br>$\sqrt{106} \cos(x + 0.507\dots)$                    |  |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> ✓</li> <li>•<sup>2</sup> ✗</li> <li>•<sup>3</sup> ✓</li> <li>•<sup>4</sup> ✓<sub>1</sub></li> </ul> |

| Question                                                    | Generic scheme                                 | Illustrative scheme                 | Max mark                                       |
|-------------------------------------------------------------|------------------------------------------------|-------------------------------------|------------------------------------------------|
| <b>6.(a) (continued)</b>                                    |                                                |                                     |                                                |
| <b>Candidate E - use of <math>x</math> at •<sup>2</sup></b> |                                                | <b>Candidate F</b>                  |                                                |
| $k \cos x \cos a - k \sin x \sin a$                         | • <sup>1</sup> ✓                               | $k \cos A \cos B - k \sin A \sin B$ | • <sup>1</sup> ✗                               |
| $k \cos x = 5$                                              |                                                | $k \cos A = 5$                      |                                                |
| $k \sin x = 9$                                              | • <sup>2</sup> ✗                               | $k \sin A = 9$                      | • <sup>2</sup> ✗                               |
| $\tan a = \frac{9}{5}$                                      |                                                | $\tan A = \frac{9}{5}$              |                                                |
| $a = 1.06\dots$                                             |                                                | $a = 1.06\dots$                     |                                                |
| $\sqrt{106} \cos(x + 1.06\dots)$                            | • <sup>3</sup> ✓ • <sup>4</sup> ✓ <sub>1</sub> | $\sqrt{106} \cos(x + 1.06\dots)$    | • <sup>3</sup> ✓ • <sup>4</sup> ✓ <sub>1</sub> |

| Question                                                                                                                                                                                                                                                                                                            |     |  | Generic scheme                                                                                                                                                                                         | Illustrative scheme                                                                                                                                                        | Max mark |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 6.                                                                                                                                                                                                                                                                                                                  | (b) |  | $\bullet^5$ link to (a)<br>$\bullet^6$ solve for $(x + a)$<br>$\bullet^7$ solve for $x$                                                                                                                | $\bullet^5 \sqrt{106} \cos(x + 1.06...) = 7$<br>$\bullet^6$ 0.82... (7.106...), $\bullet^7$ 5.46...<br>$\bullet^7$ 6.04..., 4.396...                                       | 3        |
| <b>Notes:</b><br>10. In part (b), where candidates work in degrees throughout, the maximum mark available is 2/3.<br>11. $\bullet^7$ is only available for two solutions within the stated range. Ignore 'solutions' outwith the range.<br>12. At $\bullet^7$ accept values of $x$ which round to 6.0, 6.05 or 4.4. |     |  |                                                                                                                                                                                                        |                                                                                                                                                                            |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                                                 |     |  |                                                                                                                                                                                                        |                                                                                                                                                                            |          |
| <b>Candidate G - converting to radians</b>                                                                                                                                                                                                                                                                          |     |  | $\vdots$<br>$\sqrt{106} \cos(x + 60.9...)$<br>$\sqrt{106} \cos(x + 60.9...) = 7$<br>$x + 60.9... = 312.8..., 407.1...$<br>$x = 251.8..., 346.2...$<br>$x = \frac{251.9\pi}{180}, \frac{346.2\pi}{180}$ | $\bullet^1 \checkmark \bullet^2 \checkmark \bullet^3 \checkmark$<br>$\bullet^4 \times$<br>$\bullet^5 \checkmark_1$<br>$\bullet^6 \checkmark_1$<br>$\bullet^7 \checkmark_1$ |          |
| <b>Candidate H - working in degrees</b>                                                                                                                                                                                                                                                                             |     |  | $\vdots$<br>$\sqrt{106} \cos(x + 60.9...)$<br>$\sqrt{106} \cos(x + 60.9...) = 7$<br>$x + 60.9... = 312.8..., 407.1...$<br>$x = 251.8..., 346.2...$                                                     | $\bullet^1 \checkmark \bullet^2 \checkmark \bullet^3 \checkmark$<br>$\bullet^4 \times$<br>$\bullet^5 \checkmark_1$<br>$\bullet^6 \checkmark_1 \bullet^7 \wedge$            |          |
| <b>Candidate I - working in degrees</b>                                                                                                                                                                                                                                                                             |     |  | $\vdots$<br>$\sqrt{106} \cos(x + 60.9...)$<br>$\sqrt{106} \cos(x + 60.9...) = 7$<br>$x + 60.9... = 312.8...,$<br>$x = 251.8...$                                                                        | $\bullet^1 \checkmark \bullet^2 \checkmark \bullet^3 \checkmark$<br>$\bullet^4 \times$<br>$\bullet^5 \checkmark_1$<br>$\bullet^6 \wedge \bullet^7 \wedge$                  |          |
| <b>Candidate J - working in degrees</b>                                                                                                                                                                                                                                                                             |     |  | $\vdots$<br>$\sqrt{106} \cos(x + 60.9...)$<br>$\sqrt{106} \cos(x + 60.9...) = 7$<br>$x + 60.9... = 312.8..., 407.1...$                                                                                 | $\bullet^1 \checkmark \bullet^2 \checkmark \bullet^3 \checkmark$<br>$\bullet^4 \times$<br>$\bullet^5 \checkmark_1$<br>$\bullet^6 \wedge \bullet^7 \wedge$                  |          |

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |  | Generic scheme                                                                                                                                                                                               | Illustrative scheme                                                                                                 | Max mark |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|----------|
| 7.                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |  | <p>•<sup>1</sup> start to integrate</p> <p>•<sup>2</sup> complete integration</p>                                                                                                                            | <p>•<sup>1</sup> <math>\frac{(3x+2)^8}{8}</math></p> <p>•<sup>2</sup> <math>\dots \times \frac{1}{3} + c</math></p> | 2        |
| <b>Notes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |  |                                                                                                                                                                                                              |                                                                                                                     |          |
| <p>1. Award •<sup>1</sup> for any appearance of <math>\frac{(3x+2)^8}{8}</math> regardless of any constant multiplier.</p> <p>2. Where candidates <b>differentiate throughout</b> or make <b>no attempt to integrate</b>, award 0/2.</p> <p>3. Where candidates start to integrate individual terms within the bracket or use another invalid approach, award 0/2.</p> <p>4. Do not penalise the continued appearance of the integral sign or 'dx'.</p> |  |  |                                                                                                                                                                                                              |                                                                                                                     |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                                                                                                                                                                                     |  |  |                                                                                                                                                                                                              |                                                                                                                     |          |
| <b>Candidate A</b><br>$\frac{(3x+2)^8}{8} + c$ <p>•<sup>1</sup> ✓ •<sup>2</sup> ^</p>                                                                                                                                                                                                                                                                                                                                                                   |  |  | <b>Candidate B - NOT Differentiating throughout</b><br>$7(3x+2)^6 \times \frac{1}{3} + c$ <p>•<sup>1</sup> ✗ •<sup>2</sup> ✓<sub>1</sub></p>                                                                 |                                                                                                                     |          |
| <b>Candidate C - 'Integrating' over two lines</b><br>$\frac{(3x+2)^8}{8}$ <p>•<sup>1</sup> ✓ •<sup>2</sup> ^</p> $\frac{(3x+2)^8}{8} \times \frac{1}{3} + c$ <p>SEEN</p>                                                                                                                                                                                                                                                                                |  |  | <b>Candidate D - integration incomplete at •<sup>2</sup> stage</b><br>$\frac{(3x+2)^8}{8} \times \frac{1}{3}$ <p>•<sup>1</sup> ✓ •<sup>2</sup> ^</p> $\frac{(3x+2)^8}{8} \times \frac{1}{3} + c$ <p>SEEN</p> |                                                                                                                     |          |
| <b>Candidate E - integration by substitution</b><br>$\int u^7 \times \frac{1}{3} du$ where $u = 3x+2$ and $du = 3dx$ • <sup>1</sup> ✓<br>$\frac{u^8}{24} + c$<br>$\frac{(3x+2)^8}{24} + c$ <p>•<sup>2</sup> ✓</p> <p>However, for <math>\int u^7 du</math> or <math>\int u^7 dx</math> leading to <math>\frac{(3x+2)^8}{24} + c</math> award 0/2.</p>                                                                                                   |  |  |                                                                                                                                                                                                              |                                                                                                                     |          |



| Question                                                                                                                                                                    |  |  | Generic scheme                                                                                                                                            | Illustrative scheme                                                                                                                                                                                                                                                                                                                                                    | Max mark |                                                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------------------------------------------------------------------------------|
| 8.                                                                                                                                                                          |  |  | <ul style="list-style-type: none"><li>•<sup>1</sup> identify an appropriate pathway</li><li>•<sup>2</sup> find <math>\overrightarrow{BE}</math></li></ul> | <ul style="list-style-type: none"><li>•<sup>1</sup> eg <math>\overrightarrow{BC} + \overrightarrow{CD} + \overrightarrow{DE}</math> stated or implied by •<sup>2</sup></li><li>•<sup>2</sup> <math>(0\mathbf{i} +) 5\mathbf{j} + 4\mathbf{k}</math></li></ul>                                                                                                          | 2        |                                                                                         |
| <b>Notes:</b>                                                                                                                                                               |  |  |                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                        |          |                                                                                         |
| 1. Do not penalise inconsistent vector notation (for example lack of arrows or brackets).                                                                                   |  |  |                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                        |          |                                                                                         |
| 2. • <sup>1</sup> is not available for $\overrightarrow{BD} + \overrightarrow{DE}$ or $\overrightarrow{BC} + \overrightarrow{CE}$ or similar. However, see Candidate C.     |  |  |                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                        |          |                                                                                         |
| 3. • <sup>2</sup> is only available where a valid pathway has been stated.                                                                                                  |  |  |                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                        |          |                                                                                         |
| 4. Do not accept $\begin{pmatrix} 0\mathbf{i} \\ 5\mathbf{j} \\ 4\mathbf{k} \end{pmatrix}$ or $\begin{pmatrix} 0 \\ 5 \\ 4 \end{pmatrix}$ for • <sup>2</sup> .              |  |  |                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                        |          |                                                                                         |
| 5. Where an invalid pathway ‘leads’ to $5\mathbf{j} + 4\mathbf{k}$ award 0/2.                                                                                               |  |  |                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                        |          |                                                                                         |
| <b>Commonly Observed Responses:</b>                                                                                                                                         |  |  |                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                        |          |                                                                                         |
| <b>Candidate A - using given vectors</b><br>$\overrightarrow{BE} = -\overrightarrow{DC} + \overrightarrow{AD} + \overrightarrow{DE}$<br>$5\mathbf{j} + 4\mathbf{k}$         |  |  | <ul style="list-style-type: none"><li>•<sup>1</sup> ✓</li><li>•<sup>2</sup> ✓</li></ul>                                                                   | <b>Candidate B - using given vectors</b><br>$\overrightarrow{BE} = (6\mathbf{i} + 4\mathbf{j} + 2\mathbf{k}) - (2\mathbf{i} - 4\mathbf{j} + 2\mathbf{k}) + (-4\mathbf{i} - 3\mathbf{j} + 4\mathbf{k})$<br>$\overrightarrow{BE} = \begin{pmatrix} 6 \\ 4 \\ 2 \end{pmatrix} - \begin{pmatrix} 2 \\ -4 \\ 2 \end{pmatrix} + \begin{pmatrix} -4 \\ -3 \\ 4 \end{pmatrix}$ |          | <ul style="list-style-type: none"><li>•<sup>1</sup> ✓</li><li>•<sup>1</sup> ✓</li></ul> |
| <b>Candidate C</b><br>$\overrightarrow{BD} + \overrightarrow{DE}$<br>$\overrightarrow{BD} = 4\mathbf{i} + 8\mathbf{j}$<br>$\overrightarrow{BE} = 5\mathbf{j} + 4\mathbf{k}$ |  |  | <ul style="list-style-type: none"><li>•<sup>1</sup> ✓</li><li>•<sup>2</sup> ✓</li></ul>                                                                   |                                                                                                                                                                                                                                                                                                                                                                        |          |                                                                                         |

| Question |     |  | Generic scheme                                               | Illustrative scheme                                                                                                             | Max mark |
|----------|-----|--|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|----------|
| 9.       | (a) |  | <p>•<sup>1</sup> substitute</p> <p>•<sup>2</sup> process</p> | <p>•<sup>1</sup> <math>10 = 10m + 4</math></p> <p>OR</p> $10 = \frac{4}{1-m}$ <p>•<sup>2</sup> <math>m = \frac{3}{5}</math></p> | 2        |

**Notes:**

1. Correct answer with no working, award 1/2.
2. Where candidates state ' $m = \frac{3}{5}$ ', or equivalent, and then verify the result from the given recurrence relationship, award 1/2.
3. Where candidates work in terms of  $a$ , a link to  $m$  must be made for •<sup>1</sup> to be awarded. See Candidates A, B and C.

**Commonly Observed Responses:**

|                                                                                                                                                                                                                                                                                                    |  |  |                                                                                                                                                                     |  |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
| <b>Candidate A - working not in terms of <math>m</math></b><br>(a) $10 = \frac{4}{1-a}$ • <sup>1</sup> ^<br>$a = \frac{3}{5}$ • <sup>2</sup> ✓ <sub>1</sub>                                                                                                                                        |  |  | <b>Candidate B - working not in terms of <math>m</math></b><br>(a) $10 = \frac{4}{1-a}$ • <sup>2</sup> ✓<br>$a = \frac{3}{5}$ • <sup>1</sup> ✓<br>$m = \frac{3}{5}$ |  |  |
| <b>Candidate C - working not in terms of <math>m</math></b><br>(a) $10 = \frac{4}{1-a}$ • <sup>2</sup> ✓<br>$a = \frac{3}{5}$<br>(b) $19 = \frac{3}{5}u_0 + 4$ • <sup>1</sup> ✓<br><p>•<sup>1</sup> is awarded when the calculated value of <math>a</math> is used in place of <math>m</math>.</p> |  |  |                                                                                                                                                                     |  |  |

| Question                                                                                                                                                                          |     |  | Generic scheme                 | Illustrative scheme | Max mark |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--|--------------------------------|---------------------|----------|
| 9.                                                                                                                                                                                | (b) |  | • <sup>3</sup> calculate $u_0$ | • <sup>3</sup> 52   | 1        |
| <b>Notes:</b>                                                                                                                                                                     |     |  |                                |                     |          |
| 4. Where candidates use an incorrect value of $m$ without supporting working in part (a) or which is inconsistent with their answer in part (a), • <sup>3</sup> is not available. |     |  |                                |                     |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                               |     |  |                                |                     |          |
|                                                                                                                                                                                   |     |  |                                |                     |          |

| Question |     |  | Generic scheme                                                                                                                                                                                                                                                                        | Illustrative scheme                                                                                                                                                                                                                                                                           | Max mark |
|----------|-----|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 10.      | (a) |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> express <math>P</math> in terms of <math>x</math> and <math>y</math></li> <li>•<sup>2</sup> express <math>y</math> in terms of <math>x</math></li> <li>•<sup>3</sup> substitute for <math>y</math> and complete proof</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>P = 12x + 2y</math></li> <li>•<sup>2</sup> <math>y = \frac{150 - 6x^2}{5x}</math></li> <li>•<sup>3</sup> <math>P = 12x + 2\left(\frac{150 - 6x^2}{5x}\right)</math><br/>leading to <math>P = 9.6x + \frac{60}{x}</math></li> </ul> | 3        |

**Notes:**

1. Accept  $P = 4x + 3x + y + 5x + y$  or equivalent for •<sup>1</sup>.
2. The substitution for  $y$  at •<sup>3</sup> must be clearly shown for •<sup>3</sup> to be available.
3. Do not penalise the omission of brackets at •<sup>3</sup> leading to the correct solution. See Candidate A.

**Commonly Observed Responses:**

**Candidate A - missing brackets**

∴

$$P = 12x + 2 \times \frac{150 - 6x^2}{5x}$$

$$P = 9.6x + \frac{60}{x}$$

•<sup>3</sup> ✓

| Question |     |  | Generic scheme                                                                                                                                                                                                                                                                                                                                                     | Illustrative scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Max mark |
|----------|-----|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 10.      | (b) |  | <ul style="list-style-type: none"> <li>•<sup>4</sup> express <math>P</math> in differentiable form</li> <li>•<sup>5</sup> differentiate</li> <li>•<sup>6</sup> equate expression for derivative to 0</li> <li>•<sup>7</sup> process for <math>x</math></li> <li>•<sup>8</sup> verify nature</li> <li>•<sup>9</sup> find minimum value of <math>P</math></li> </ul> | <ul style="list-style-type: none"> <li>•<sup>4</sup> <math>9.6x + 60x^{-1}</math> stated or implied by •<sup>5</sup></li> <li>•<sup>5</sup> <math>9.6 - 60x^{-2}</math></li> <li>•<sup>6</sup> <math>9.6 - 60x^{-2} = 0</math></li> <li>•<sup>7</sup> 2.5 or <math>\frac{5}{2}</math></li> <li>•<sup>8</sup> table of signs for a derivative<br/><math>\therefore</math> minimum</li> <li>OR</li> <li><math>P''(x) = 120x^{-3}</math> and<br/><math>P''(2.5) = 7.68... &gt; 0 \therefore</math> minimum</li> <li>•<sup>9</sup> 48(cm)</li> </ul> | 6        |

#### Notes:

4. For a numerical approach, award 0/6.
5. •<sup>6</sup> can be awarded for  $60x^{-2} = 9.6$ .
6. Where candidates integrate any term at the •<sup>5</sup> stage, only •<sup>6</sup> is available on follow through for setting their 'derivative' to 0.
7. •<sup>7</sup>, •<sup>8</sup> and •<sup>9</sup> are only available for working with a derivative which contains an index  $\leq -2$ .
8.  $\left(\frac{60}{9.6}\right)^{0.5}$  or  $\sqrt[2]{\frac{9.6}{60}}$  must be simplified at •<sup>7</sup> or •<sup>8</sup> for •<sup>7</sup> to be awarded.
9. Ignore the appearance of -2.5 at •<sup>7</sup>.
10. Notation for the derivative is only assessed at •<sup>8</sup>.
11. •<sup>8</sup> is not available to candidates who consider a value of  $x \leq 0$  in the neighbourhood of 2.5.
12. •<sup>9</sup> is still available in cases where a candidate's table of signs does not lead legitimately to a minimum at •<sup>8</sup>.
13. •<sup>8</sup> and •<sup>9</sup> are not available to candidates who state that the minimum exists at a value of  $x$  where  $x \leq 0$ .

#### Commonly Observed Responses:

|                                                                                                                                                                                                          |                                                                                                                                                                                                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Candidate B - differentiating over multiple lines</b><br>$P'(x) = 9.6 + 60x^{-1}$ • <sup>4</sup> ✗<br>$P'(x) = 9.6 - 60x^{-2}$ • <sup>5</sup> ✗<br>$9.6 - 60x^{-2} = 0$ • <sup>6</sup> ✓ <sub>1</sub> | <b>Candidate C - differentiating over multiple lines</b><br>$P'(x) = 9.6x + 60x^{-1}$ • <sup>4</sup> ✓<br>$P'(x) = 9.6 - 60x^{-2}$ • <sup>5</sup> ✗<br>$9.6 - 60x^{-2} = 0$ • <sup>6</sup> ✓ <sub>1</sub> |
| <b>Candidate D</b><br>Stationary points when $P'(x) = 0$<br>$P'(x) = 9.6 - 60x^{-2}$ • <sup>4</sup> ✓   • <sup>5</sup> ✓   • <sup>6</sup> ✓                                                              |                                                                                                                                                                                                           |

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Generic scheme | Illustrative scheme | Max mark      |               |               |               |         |     |                |             |       |           |                                                                                                                                                                                                                                                                                                                                                                                                                             |           |               |             |               |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |     |                |             |       |           |                                                                                                                                                                                                                                                                                                                                                                                                                   |     |     |       |     |         |                |           |       |                |             |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|---------------------|---------------|---------------|---------------|---------------|---------|-----|----------------|-------------|-------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|-------------|---------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|----------------|-------------|-------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-------|-----|---------|----------------|-----------|-------|----------------|-------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| 10.(b) (continued)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                |                     |               |               |               |               |         |     |                |             |       |           |                                                                                                                                                                                                                                                                                                                                                                                                                             |           |               |             |               |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |     |                |             |       |           |                                                                                                                                                                                                                                                                                                                                                                                                                   |     |     |       |     |         |                |           |       |                |             |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |
| For the table of signs for a derivative, <b>accept</b> :                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                |                     |               |               |               |               |         |     |                |             |       |           |                                                                                                                                                                                                                                                                                                                                                                                                                             |           |               |             |               |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |     |                |             |       |           |                                                                                                                                                                                                                                                                                                                                                                                                                   |     |     |       |     |         |                |           |       |                |             |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |
| <table><tr><td><math>x</math></td><td><math>2.5^-</math></td><td><math>2.5</math></td><td><math>2.5^+</math></td></tr><tr><td><math>P'(x)</math></td><td><math>-</math></td><td><math>0</math></td><td><math>+</math></td></tr><tr><td>Slope or shape</td><td><math>\diagdown</math></td><td>_____</td><td><math>\diagup</math></td></tr></table>                                                                                                                                                                                                                                                                    | $x$            | $2.5^-$             | $2.5$         | $2.5^+$       | $P'(x)$       | $-$           | $0$     | $+$ | Slope or shape | $\diagdown$ | _____ | $\diagup$ | <table><tr><td><math>x</math></td><td><math>\rightarrow</math></td><td><math>2.5</math></td><td><math>\rightarrow</math></td></tr><tr><td><math>P'(x)</math></td><td><math>-</math></td><td><math>0</math></td><td><math>+</math></td></tr><tr><td>Slope or shape</td><td><math>\diagdown</math></td><td>_____</td><td><math>\diagup</math></td></tr></table> <p>Arrows are taken to mean<br/>'in the neighbourhood of'</p> | $x$       | $\rightarrow$ | $2.5$       | $\rightarrow$ | $P'(x)$   | $-$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $0$ | $+$ | Slope or shape | $\diagdown$ | _____ | $\diagup$ | <table><tr><td><math>x</math></td><td><math>a</math></td><td><math>2.5</math></td><td><math>b</math></td></tr><tr><td><math>P'(x)</math></td><td><math>-</math></td><td><math>0</math></td><td><math>+</math></td></tr><tr><td>Slope or shape</td><td><math>\diagdown</math></td><td>_____</td><td><math>\diagup</math></td></tr></table> <p>Where <math>0 &lt; a &lt; 2.5</math> and <math>b &gt; 2.5</math></p> | $x$ | $a$ | $2.5$ | $b$ | $P'(x)$ | $-$            | $0$       | $+$   | Slope or shape | $\diagdown$ | _____     | $\diagup$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |
| $x$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $2.5^-$        | $2.5$               | $2.5^+$       |               |               |               |         |     |                |             |       |           |                                                                                                                                                                                                                                                                                                                                                                                                                             |           |               |             |               |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |     |                |             |       |           |                                                                                                                                                                                                                                                                                                                                                                                                                   |     |     |       |     |         |                |           |       |                |             |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |
| $P'(x)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $-$            | $0$                 | $+$           |               |               |               |         |     |                |             |       |           |                                                                                                                                                                                                                                                                                                                                                                                                                             |           |               |             |               |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |     |                |             |       |           |                                                                                                                                                                                                                                                                                                                                                                                                                   |     |     |       |     |         |                |           |       |                |             |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |
| Slope or shape                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $\diagdown$    | _____               | $\diagup$     |               |               |               |         |     |                |             |       |           |                                                                                                                                                                                                                                                                                                                                                                                                                             |           |               |             |               |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |     |                |             |       |           |                                                                                                                                                                                                                                                                                                                                                                                                                   |     |     |       |     |         |                |           |       |                |             |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |
| $x$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $\rightarrow$  | $2.5$               | $\rightarrow$ |               |               |               |         |     |                |             |       |           |                                                                                                                                                                                                                                                                                                                                                                                                                             |           |               |             |               |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |     |                |             |       |           |                                                                                                                                                                                                                                                                                                                                                                                                                   |     |     |       |     |         |                |           |       |                |             |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |
| $P'(x)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $-$            | $0$                 | $+$           |               |               |               |         |     |                |             |       |           |                                                                                                                                                                                                                                                                                                                                                                                                                             |           |               |             |               |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |     |                |             |       |           |                                                                                                                                                                                                                                                                                                                                                                                                                   |     |     |       |     |         |                |           |       |                |             |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |
| Slope or shape                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $\diagdown$    | _____               | $\diagup$     |               |               |               |         |     |                |             |       |           |                                                                                                                                                                                                                                                                                                                                                                                                                             |           |               |             |               |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |     |                |             |       |           |                                                                                                                                                                                                                                                                                                                                                                                                                   |     |     |       |     |         |                |           |       |                |             |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |
| $x$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $a$            | $2.5$               | $b$           |               |               |               |         |     |                |             |       |           |                                                                                                                                                                                                                                                                                                                                                                                                                             |           |               |             |               |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |     |                |             |       |           |                                                                                                                                                                                                                                                                                                                                                                                                                   |     |     |       |     |         |                |           |       |                |             |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |
| $P'(x)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $-$            | $0$                 | $+$           |               |               |               |         |     |                |             |       |           |                                                                                                                                                                                                                                                                                                                                                                                                                             |           |               |             |               |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |     |                |             |       |           |                                                                                                                                                                                                                                                                                                                                                                                                                   |     |     |       |     |         |                |           |       |                |             |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |
| Slope or shape                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $\diagdown$    | _____               | $\diagup$     |               |               |               |         |     |                |             |       |           |                                                                                                                                                                                                                                                                                                                                                                                                                             |           |               |             |               |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |     |                |             |       |           |                                                                                                                                                                                                                                                                                                                                                                                                                   |     |     |       |     |         |                |           |       |                |             |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |
| For the table of signs for a derivative, do <b>NOT</b> accept:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                |                     |               |               |               |               |         |     |                |             |       |           |                                                                                                                                                                                                                                                                                                                                                                                                                             |           |               |             |               |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |     |                |             |       |           |                                                                                                                                                                                                                                                                                                                                                                                                                   |     |     |       |     |         |                |           |       |                |             |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |
| <table><tr><td><math>x</math></td><td><math>\rightarrow</math></td><td><math>-2.5</math></td><td><math>\rightarrow</math></td><td><math>2.5</math></td><td><math>\rightarrow</math></td></tr><tr><td><math>P'(x)</math></td><td><math>+</math></td><td><math>0</math></td><td><math>-</math></td><td><math>0</math></td><td><math>+</math></td></tr><tr><td>Slope or shape</td><td><math>\diagup</math></td><td>_____</td><td><math>\diagdown</math></td><td>_____</td><td><math>\diagup</math></td></tr></table> <p>Since the function is discontinuous<br/><math>-2.5 \rightarrow 2.5</math> is NOT acceptable</p> | $x$            | $\rightarrow$       | $-2.5$        | $\rightarrow$ | $2.5$         | $\rightarrow$ | $P'(x)$ | $+$ | $0$            | $-$         | $0$   | $+$       | Slope or shape                                                                                                                                                                                                                                                                                                                                                                                                              | $\diagup$ | _____         | $\diagdown$ | _____         | $\diagup$ | <table><tr><td><math>x</math></td><td><math>a</math></td><td><math>-2.5</math></td><td><math>b</math></td><td><math>2.5</math></td><td><math>c</math></td></tr><tr><td><math>P'(x)</math></td><td><math>+</math></td><td><math>0</math></td><td><math>-</math></td><td><math>0</math></td><td><math>+</math></td></tr><tr><td>Slope or shape</td><td><math>\diagup</math></td><td>_____</td><td><math>\diagdown</math></td><td>_____</td><td><math>\diagup</math></td></tr></table> <p>Since the function is discontinuous<br/><math>-2.5 &lt; b &lt; 2.5</math> is NOT acceptable</p> | $x$ | $a$ | $-2.5$         | $b$         | $2.5$ | $c$       | $P'(x)$                                                                                                                                                                                                                                                                                                                                                                                                           | $+$ | $0$ | $-$   | $0$ | $+$     | Slope or shape | $\diagup$ | _____ | $\diagdown$    | _____       | $\diagup$ | <ul style="list-style-type: none"><li>For this question, do not penalise the omission of '<math>x</math>' or the word 'shape'/'slope'</li><li>Stating values of <math>P'(x)</math> is an acceptable alternative to writing '+' or '-' signs</li><li>Acceptable variations of <math>P'(x)</math> are: <math>P'</math>, <math>\frac{dP}{dx}</math>, and <math>9.6 - 60x^{-2}</math>. Accept <math>f'(x)</math> only where candidates have previously used <math>f(x) = 9.6x + 60x^{-1}</math> in their working</li><li>Do not accept <math>\frac{dy}{dx}</math> or <math>\frac{d^2y}{dx^2}</math></li></ul> |  |
| $x$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $\rightarrow$  | $-2.5$              | $\rightarrow$ | $2.5$         | $\rightarrow$ |               |         |     |                |             |       |           |                                                                                                                                                                                                                                                                                                                                                                                                                             |           |               |             |               |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |     |                |             |       |           |                                                                                                                                                                                                                                                                                                                                                                                                                   |     |     |       |     |         |                |           |       |                |             |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |
| $P'(x)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $+$            | $0$                 | $-$           | $0$           | $+$           |               |         |     |                |             |       |           |                                                                                                                                                                                                                                                                                                                                                                                                                             |           |               |             |               |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |     |                |             |       |           |                                                                                                                                                                                                                                                                                                                                                                                                                   |     |     |       |     |         |                |           |       |                |             |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |
| Slope or shape                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $\diagup$      | _____               | $\diagdown$   | _____         | $\diagup$     |               |         |     |                |             |       |           |                                                                                                                                                                                                                                                                                                                                                                                                                             |           |               |             |               |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |     |                |             |       |           |                                                                                                                                                                                                                                                                                                                                                                                                                   |     |     |       |     |         |                |           |       |                |             |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |
| $x$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $a$            | $-2.5$              | $b$           | $2.5$         | $c$           |               |         |     |                |             |       |           |                                                                                                                                                                                                                                                                                                                                                                                                                             |           |               |             |               |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |     |                |             |       |           |                                                                                                                                                                                                                                                                                                                                                                                                                   |     |     |       |     |         |                |           |       |                |             |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |
| $P'(x)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $+$            | $0$                 | $-$           | $0$           | $+$           |               |         |     |                |             |       |           |                                                                                                                                                                                                                                                                                                                                                                                                                             |           |               |             |               |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |     |                |             |       |           |                                                                                                                                                                                                                                                                                                                                                                                                                   |     |     |       |     |         |                |           |       |                |             |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |
| Slope or shape                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $\diagup$      | _____               | $\diagdown$   | _____         | $\diagup$     |               |         |     |                |             |       |           |                                                                                                                                                                                                                                                                                                                                                                                                                             |           |               |             |               |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |     |                |             |       |           |                                                                                                                                                                                                                                                                                                                                                                                                                   |     |     |       |     |         |                |           |       |                |             |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |  | Generic scheme                                                                                                                                                                                                                                                                                           | Illustrative scheme                                                                                                                                                                                                                                                                                                                                                 | Max mark |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 11.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> substitute double angle formula for <math>\sin 2x^\circ</math></li> <li>•<sup>2</sup> factorise</li> <li>•<sup>3</sup> solve for <math>\cos x^\circ</math> and <math>\sin x^\circ</math></li> <li>•<sup>4</sup> solve for <math>x</math></li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>3(2 \sin x^\circ \cos x^\circ) + 4 \cos x^\circ (= 0)</math></li> <li>•<sup>2</sup> eg <math>2 \cos x^\circ (3 \sin x^\circ + 2) = 0</math></li> <li>•<sup>3</sup> <math>\cos x^\circ = 0</math>, <math>\sin x^\circ = -\frac{2}{3}</math></li> <li>•<sup>4</sup> 90, 270, 221.8..., 318.1...</li> </ul> | 4        |
| <b>Notes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |  |                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                     |          |
| <p>1. Substituting <math>2 \sin A \cos A</math> for <math>\sin 2x^\circ</math> at the •<sup>1</sup> stage should be treated as bad form provided the equation is written in terms of <math>x</math> at the •<sup>2</sup> stage. Otherwise, •<sup>1</sup> is not available.</p> <p>2. ' = 0 ' must appear by the •<sup>2</sup> stage for •<sup>2</sup> to be awarded.</p> <p>3. Do not penalise the absence of '2' as a common factor at •<sup>2</sup>.</p> <p>4. Award •<sup>3</sup> for <math>x = \cos^{-1}(0)</math> <b>AND</b> <math>x = \sin^{-1}\left(-\frac{2}{3}\right)</math>.</p> <p>5. Do not penalise the omission of degree signs.</p> <p>6. Candidates who leave their answer in radians do not gain •<sup>4</sup> (if marking horizontally) or •<sup>3</sup> (if marking vertically).</p> <p>7. Where equations for <math>\sin x^\circ</math> and/or <math>\cos x^\circ</math> do not have solutions, marks are unavailable for stating 'no solutions'.</p> |  |  |                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                     |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                     |          |
| <b>Candidate A - dividing by <math>\cos x</math></b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |  | <b>Candidate B - insufficient evidence for •<sup>3</sup></b>                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                     |          |
| $6 \sin x^\circ \cos x^\circ = -4 \cos x^\circ$<br>$6 \sin x^\circ = -4$<br>$x = 221.8..., 318.1...$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |  | $6 \sin x^\circ \cos x^\circ + 4 \cos x^\circ = 0$<br>$2 \cos x^\circ (3 \sin x^\circ + 2) = 0$<br>$2 \cos x^\circ = 0, \sin x^\circ = -\frac{2}{3}$<br>However,<br>$x = 90, 270, 221.8..., 318.1...$                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                     |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |  |                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                     |          |
| <b>Candidate C</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |  |                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                     |          |
| $\cos x^\circ = 0$ , $\sin x^\circ = -\frac{2}{3}$<br>$x = 90, 270$ $x = 41$<br>$x = 221.8..., 318.1...$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |  |                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                     |          |
| However, where the final solution(s) are clearly identified by the candidate award • <sup>4</sup> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                     |          |

| Question                                                                                                 |     |  | Generic scheme                                                                                                            | Illustrative scheme                                                                                                                   | Max mark |
|----------------------------------------------------------------------------------------------------------|-----|--|---------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|----------|
| 12.                                                                                                      | (a) |  | <p>•<sup>1</sup> interpret notation</p> <p>•<sup>2</sup> state expression for <math>f(g(x))</math></p>                    | <p>•<sup>1</sup> <math>f(1-x^3)</math></p> <p>OR</p> <p><math>(g(x))^5 + 3</math></p> <p>•<sup>2</sup> <math>(1-x^3)^5 + 3</math></p> | 2        |
| <b>Notes:</b>                                                                                            |     |  |                                                                                                                           |                                                                                                                                       |          |
| 1. For $(1-x^3)^5 + 3$ without working, award 2/2.                                                       |     |  |                                                                                                                           |                                                                                                                                       |          |
| <b>Commonly Observed Responses:</b>                                                                      |     |  |                                                                                                                           |                                                                                                                                       |          |
| <b>Candidate A</b><br>$f(g(x)) = (1-x^3)^5 + 3$ • <sup>1</sup> ✓ • <sup>2</sup> ✓<br>$h(x) = 4 - x^{15}$ |     |  | <b>Candidate B - two 'attempts'</b><br>$f(g(x)) = x^5 + 3$ • <sup>1</sup> ✗ • <sup>2</sup> ✗<br>$f(g(x)) = (1-x^3)^5 + 3$ |                                                                                                                                       |          |
| <b>Candidate C</b><br>$f(g(x)) = 1 - (x^5 + 3)^3$ • <sup>1</sup> ✗ • <sup>2</sup> ✓ <sub>1</sub>         |     |  |                                                                                                                           |                                                                                                                                       |          |





| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |     |  | Generic scheme                                                                                                                                                    | Illustrative scheme                                                                                                                                                                                                                                  | Max mark |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 13.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | (a) |  | • <sup>1</sup> identify initial mass                                                                                                                              | • <sup>1</sup> 150 (micrograms)                                                                                                                                                                                                                      | 1        |
| Notes:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |  |                                                                                                                                                                   |                                                                                                                                                                                                                                                      |          |
| Commonly Observed Responses:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |     |  |                                                                                                                                                                   |                                                                                                                                                                                                                                                      |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | (b) |  | • <sup>2</sup> interpret information<br><br>• <sup>3</sup> process equation<br><br>• <sup>4</sup> write in logarithmic form<br><br>• <sup>5</sup> process for $t$ | • <sup>2</sup> $120 = 150e^{-0.0054t}$<br>stated or implied by • <sup>3</sup><br><br>• <sup>3</sup> $e^{-0.0054t} = \frac{120}{150}$<br><br>• <sup>4</sup> $\log_e \left( \frac{120}{150} \right) = -0.0054t$<br><br>• <sup>5</sup> 41.32... (years) | 4        |
| Notes:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |  |                                                                                                                                                                   |                                                                                                                                                                                                                                                      |          |
| 1. Where values other than 120 are used in the substitution, • <sup>3</sup> , • <sup>4</sup> and • <sup>5</sup> are still available.<br>2. • <sup>3</sup> may be implied by • <sup>4</sup> .<br>3. Evidence for • <sup>4</sup> must be stated explicitly. See Candidate B.<br>4. At • <sup>4</sup> all exponentials must be processed.<br>5. Any base may be used at • <sup>4</sup> stage. See Candidate A.<br>6. Accept $\ln 0.8 = -0.0054t \ln e$ and $-0.223... = -0.0054t$ for • <sup>4</sup> .<br>7. • <sup>5</sup> is unavailable where candidates round the value of $\ln 0.8$ to fewer than 2 decimal places.<br>8. Accept answers where $40.7 \leq t \leq 42$ at • <sup>5</sup> .<br>9. The calculation at • <sup>5</sup> must follow from the valid use of exponentials and logarithms at • <sup>3</sup> and • <sup>4</sup> .<br>10. Where candidates show no working or take an iterative approach to arrive at $t = 41$ or $t = 42$ , award 1/4. However, if, in any iterations $M$ is evaluated for $t = 41$ and $t = 42$ leading to a final answer of $t = 42$ (years), then award 4/4. |     |  |                                                                                                                                                                   |                                                                                                                                                                                                                                                      |          |
| Commonly Observed Responses:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |     |  |                                                                                                                                                                   |                                                                                                                                                                                                                                                      |          |
| Candidate A - using other bases                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |  | Candidate B - missing working                                                                                                                                     |                                                                                                                                                                                                                                                      |          |
| $120 = 150e^{-0.0054t}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |  | $120 = 150e^{-0.0054t}$                                                                                                                                           |                                                                                                                                                                                                                                                      |          |
| $0.8 = e^{-0.0054t}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |     |  | $0.8 = e^{-0.0054t}$                                                                                                                                              |                                                                                                                                                                                                                                                      |          |
| $\log_{10} 0.8 = -0.0054t \log_{10} e$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |  | $t = 41$                                                                                                                                                          |                                                                                                                                                                                                                                                      |          |
| $t = 41.32...$ (years)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |  |                                                                                                                                                                   |                                                                                                                                                                                                                                                      |          |
| Candidate C - iterative approach                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |     |  | Candidate D - taking logarithms of both sides                                                                                                                     |                                                                                                                                                                                                                                                      |          |
| $t = 40 \Rightarrow M = 120.86...$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |  | $120 = 150e^{-0.0054t}$                                                                                                                                           |                                                                                                                                                                                                                                                      |          |
| $t = 41 \Rightarrow M = 120.20...$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |  | $\log_e 120 = \log_e (150e^{-0.0054t})$                                                                                                                           |                                                                                                                                                                                                                                                      |          |
| $t = 42 \Rightarrow M = 119.56...$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |  | $\log_e 120 = \log_e 150 + \log_e e^{-0.0054t}$                                                                                                                   |                                                                                                                                                                                                                                                      |          |
| So $t = 42$ (years)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |  | $\log_e 120 - \log_e 150 = -0.0054t$                                                                                                                              |                                                                                                                                                                                                                                                      |          |
| Award 4/4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |  | $t = 41.32...$ (years)                                                                                                                                            |                                                                                                                                                                                                                                                      |          |

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                   |      |                               | Generic scheme                                                                                                                                                                                                                                                                                   | Illustrative scheme                                                                                                                                                                        | Max mark         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 14.                                                                                                                                                                                                                                                                                                                                                                                                                                                        | (a)  |                               | <ul style="list-style-type: none"> <li>•<sup>1</sup> state coordinates of centre</li> <li>•<sup>2</sup> state radius</li> </ul>                                                                                                                                                                  | <ul style="list-style-type: none"> <li>•<sup>1</sup> (−5,6)</li> <li>•<sup>2</sup> 3</li> </ul>                                                                                            | 2                |
| <b>Notes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                              |      |                               |                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                            |                  |
| 1. Accept $x = -5$ , $y = 6$ for • <sup>1</sup> .<br>2. Do not accept ' $a = \dots, b = \dots$ ' or '−5,6' for • <sup>1</sup> .                                                                                                                                                                                                                                                                                                                            |      |                               |                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                            |                  |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                                                                                                                                                                                        |      |                               |                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                            |                  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                            | (b)  |                               | <ul style="list-style-type: none"> <li>•<sup>3</sup> find coordinates of centre</li> <li>•<sup>4</sup> find radius</li> </ul>                                                                                                                                                                    | <ul style="list-style-type: none"> <li>•<sup>3</sup> (7,−3)</li> <li>•<sup>4</sup> 2</li> </ul>                                                                                            | 2                |
| <b>Notes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                              |      |                               |                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                            |                  |
| 3. Accept $x = 7$ , $y = -3$ for • <sup>3</sup> .<br>4. Do not accept ' $g = \dots, f = \dots$ ' or '7,−3' for • <sup>3</sup> .<br>5. Do not penalise candidates who treat negatives with a lack of rigour when calculating the radius.<br>For example, accept $\sqrt{7^2 + 3^2 - 54} = 2$ or $\sqrt{7^2 + -3^2 - 54} = 2$ or $\sqrt{-7^2 + 3^2 - 54} = 2$ for • <sup>4</sup> .<br>However, do not accept $\sqrt{7^2 - 3^2 - 54} = 2$ for • <sup>4</sup> . |      |                               |                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                            |                  |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                                                                                                                                                                                        |      |                               |                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                            |                  |
| <b>Candidate A - repeated error within a question</b>                                                                                                                                                                                                                                                                                                                                                                                                      |      |                               | <b>Candidate B - two errors</b>                                                                                                                                                                                                                                                                  |                                                                                                                                                                                            |                  |
| (a)                                                                                                                                                                                                                                                                                                                                                                                                                                                        | −5,6 | • <sup>1</sup> ✗              | (a)                                                                                                                                                                                                                                                                                              | 5,−6                                                                                                                                                                                       | • <sup>1</sup> ✗ |
| (b)                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 7,−3 | • <sup>3</sup> ✓ <sub>1</sub> | (b)                                                                                                                                                                                                                                                                                              | 7,−3                                                                                                                                                                                       | • <sup>3</sup> ✗ |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                            | (c)  |                               | <ul style="list-style-type: none"> <li>•<sup>5</sup> find distance between centres of <math>C_1</math> and <math>C_2</math></li> <li>•<sup>6</sup> calculate radius of <math>C_3</math></li> <li>•<sup>7</sup> find centre of <math>C_3</math> and state equation of <math>C_3</math></li> </ul> | <ul style="list-style-type: none"> <li>•<sup>5</sup> 15</li> <li>•<sup>6</sup> 8</li> <li>•<sup>7</sup> (−1,3) <b>stated explicitly</b> and <math>(x+1)^2 + (y-3)^2 = 64</math></li> </ul> | 3                |
| <b>Notes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                              |      |                               |                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                            |                  |
| 6. • <sup>5</sup> may be awarded for $\sqrt{(-5-7)^2 + (6+3)^2}$ within a valid calculation for the radius of $C_3$ .<br>7. • <sup>6</sup> is only available where a valid approach to finding the distance between the centres of $C_1$ and $C_2$ has been used.<br>8. Where candidates use a radius without valid supporting working, • <sup>7</sup> is not available.<br>9. Accept the centre of $C_3$ written as a position vector.                    |      |                               |                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                            |                  |

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Generic scheme | Illustrative scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Max mark |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>14.(c) (continued)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |          |
| <b>Valid approaches for finding <math>r</math></b><br><br>$r = \frac{15+3-2}{2} \therefore r = 8$ <span style="float: right;">•<sup>6</sup> ✓</span><br><br>$r = \frac{\sqrt{(7+5)^2 + (6+3)^2} + 3 - 2}{2} \therefore r = 8$ <span style="float: right;">•<sup>5</sup> ✓ •<sup>6</sup> ✓</span><br><br>$r = 3 + 3 + \frac{(13-9)}{2} \therefore r = 8$ <span style="float: right;">•<sup>6</sup> ✓</span><br><br>Let $x = r_3 - r_1$ . Since $r_3 = 13 - x$ ,<br>$x = (13 - x) - 3 \Rightarrow x = 5 \therefore r = 8$ <span style="float: right;">•<sup>6</sup> ✓</span><br>(may also use $y + 3 = r_3 - r_1$ )<br><br><b>This list is not exhaustive</b>        |                | <b>Invalid approaches and/or insufficient communication for finding <math>r</math></b><br><br>$r = 3 + 3 + 2 \therefore r = 8$ <span style="float: right;">•<sup>6</sup> ^</span><br><br>gap = 2 $\therefore r = 10 - 2 \therefore r = 8$ <span style="float: right;">•<sup>6</sup> ^</span><br><br>$r = \frac{15}{3} + 3 \therefore r = 8$ <span style="float: right;">•<sup>6</sup> ^</span><br><br>Distance = 15 $\therefore r = 8$<br>(with no supporting working) <span style="float: right;">•<sup>6</sup> ^</span><br><br><b>This list is not exhaustive</b> |          |
| <b>Valid approaches for finding centre of <math>C_3</math></b><br><br>Ratio = 1:2 $\therefore C_3(-1,3)$<br><br>$C_3\left(-5 + 12 \times \frac{1}{3}, 6 - 9 \times \frac{1}{3}\right) \therefore C_3(-1,3)$<br><br>Ratio = 15:10 & stepping out $\therefore C_3(-1,3)$<br>(may be drawn as similar triangles)<br><br>Let $x = r_3 - r_1$ . Since $r_3 = 13 - x$ ,<br>$x = (13 - x) - 3 \Rightarrow x = 5$<br>$\Rightarrow$ Ratio = 5:10 $\therefore C_3(-1,3)$<br>(may also use $y + 3 = r_3 - r_1$ )<br><br>$2\overrightarrow{C_1C_3} = \overrightarrow{C_3C_2} \therefore C_3 = \begin{pmatrix} -1 \\ 3 \end{pmatrix}$<br><br><b>This list is not exhaustive</b> |                | <b>Invalid approaches and/or insufficient communication for finding centre of <math>C_3</math></b><br><br>$m_{C_1C_2} = -\frac{3}{4} \therefore C_1(-5,6) \rightarrow C_3(-1,3)$ <span style="float: right;">•<sup>7</sup> ✗</span><br><br>$\frac{y-6}{x+5} = \frac{-3-y}{7-x} = -\frac{3}{4} \therefore C_3(-1,3)$ <span style="float: right;">•<sup>7</sup> ✗</span><br><br>{3,4,5} triangle with no evidence<br>of a ratio $\therefore C_3(-1,3)$ <span style="float: right;">•<sup>7</sup> ✗</span><br><br><b>This list is not exhaustive</b>                   |          |

[END OF MARKING INSTRUCTIONS]